Day 088 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Convert Metric Units with Ratios

Standards: 6.RP.A.3d

Learning target: I can convert measurements using equivalent ratios.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

Day 088 | Unit 06 | Lesson

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01 / Learn

Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

Worked example 1

Convert 5 liters to milliliters. Show the conversion ratio.

1. Write an equality such as 1 foot = 12 inches.

2. Multiply to change larger units into smaller units; divide for the reverse.

3. 5 × 1000 = 5000.

4. Smaller units require more units for the same amount.

5. State the result in the form requested: 5000 milliliters.

Answer: 5000 milliliters

Worked example 2

Use 1 meters = 100 centimeters. Select all statements equivalent to 11/2 meters.
A. 551 centimeters
B. 550 centimeters
C. 11/2 × 100 centimeters
D. 11/200 centimeters

1. Write an equality such as 1 foot = 12 inches.

2. There are 100 smaller units in each larger one.

3. Multiply 11/2 by 100, giving 550 centimeters.

4. The physical length does not change.

Answer: B, C - 550 centimeters; 11/2 × 100 centimeters

Day 088 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Convert 7 liters to milliliters. Show the conversion ratio.

02 / Constructed response

Convert 3 liters to milliliters. Show the conversion ratio.

03 / Constructed response

Convert 5 meters to centimeters. Show the conversion ratio.

04 / Constructed response

Convert 8000 milliliters to liters. Show the conversion ratio.

05 / Constructed response

Convert 7 meters to centimeters. Show the conversion ratio.

06 / Constructed response

Convert 600 centimeters to meters. Show the conversion ratio.

07 / Constructed response

Convert 3 meters to centimeters. Show the conversion ratio and explain why the numerical value grows.

08 / Multiple select

Use 1 meters = 100 centimeters. Select all statements equivalent to 13/2 meters.
A. 651 centimeters
B. 650 centimeters
C. 13/200 centimeters
D. 13/2 × 100 centimeters

09 / Multiple select

Use 1 meters = 100 centimeters. Select all statements equivalent to 7/2 meters.
A. 351 centimeters
B. 7/200 centimeters
C. 7/2 × 100 centimeters
D. 350 centimeters

10 / Constructed response

Convert 2000 milliliters to liters. Show the conversion ratio.

Day 088 | Unit 06 | Answer key

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1. 7000 milliliters

7 × 1000 = 7000. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

2. 3000 milliliters

3 × 1000 = 3000. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

3. 500 centimeters

5 × 100 = 500. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

4. 8 liters

8000 ÷ 1000 = 8; 1 liters = 1000 milliliters.

6.RP.A.3d | Constructed response

5. 700 centimeters

7 × 100 = 700. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

6. 6 meters

600 ÷ 100 = 6; 1 meters = 100 centimeters.

6.RP.A.3d | Constructed response

7. 300 centimeters

3 × 100 = 300. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

8. B, D - 650 centimeters; 13/2 × 100 centimeters

There are 100 smaller units in each larger one. Multiply 13/2 by 100, giving 650 centimeters. The physical length does not change.

6.RP.A.3d | Multiple select

9. C, D - 7/2 × 100 centimeters; 350 centimeters

There are 100 smaller units in each larger one. Multiply 7/2 by 100, giving 350 centimeters. The physical length does not change.

6.RP.A.3d | Multiple select

10. 2 liters

2000 ÷ 1000 = 2; 1 liters = 1000 milliliters.

6.RP.A.3d | Constructed response